

Delta Robot — Axis & Transform Reference

Coordinate frames · Camera transform · Image zones · Tuning parameters

1 Robot Base Frame & Belt Zones

All robot coordinates use the **base frame**: origin at robot arm mount, **+Y toward belt entry (approach)**, **+X right**, **+Z up**. Belt surface is at **Z = -410 mm**. Workspace bounds: $|X| \leq 150$ mm, $|Y| \leq 150$ mm.

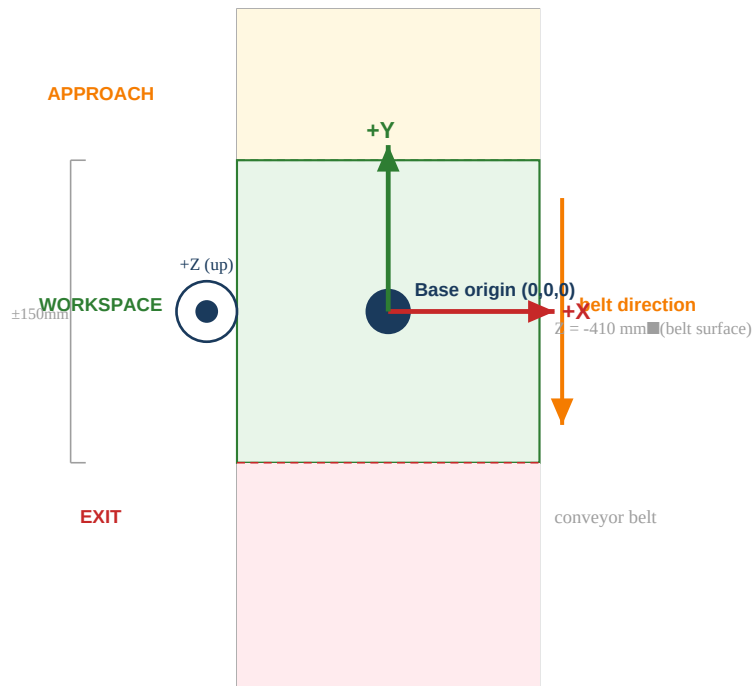


Fig 1 — Top view: robot base frame & belt zones ($Y_LIMIT = \pm 150$ mm, $X_LIMIT = \pm 150$ mm)

2 Camera Frame & Transform

The camera is mounted above the belt looking down. **CAMERA_DIRECT_MATRIX** maps camera axes to base axes. **CAM_TY_MM = -300** means the camera is 300 mm toward the EXIT side from the base origin.

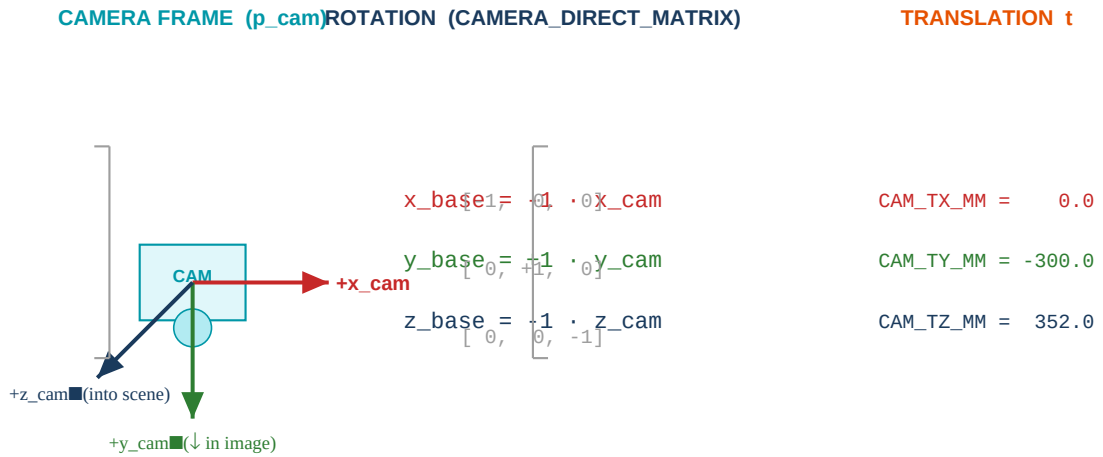


Fig 2 — Camera frame axes, rotation matrix, and translation vector

Transform equation ($T_{cam_to_base}$)

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p_base = R @ p_cam + t
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$$\begin{aligned} x_{base} &= -x_{cam} + CAM_TX_MM & (0.0) \\ y_{base} &= +y_{cam} + CAM_TY_MM & (-300.0 \text{ mm}) \\ z_{base} &= -z_{cam} + CAM_TZ_MM & (352.0 \text{ mm}) \end{aligned}$$

3 Image Frame & Zone Mapping

Pixel v **increases downward**. Approach zone (high $+y_{base}$) appears at the **bottom** of the image; exit zone (negative y_{base}) at the **top**. The workspace overlay is projected from base frame corners ($\pm X_LIMIT$, $\pm Y_LIMIT$) at $WORKSPACE_PICK_Z_MM$.

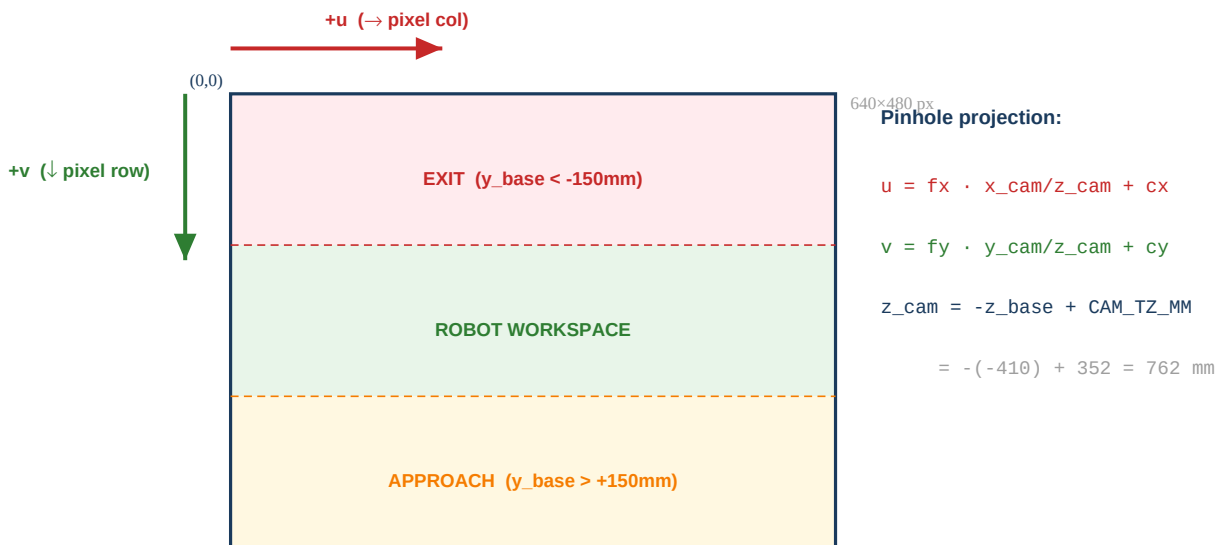


Fig 3 — Image pixel frame and zone mapping (exit=top, approach=bottom)

4 Full Transform Chain

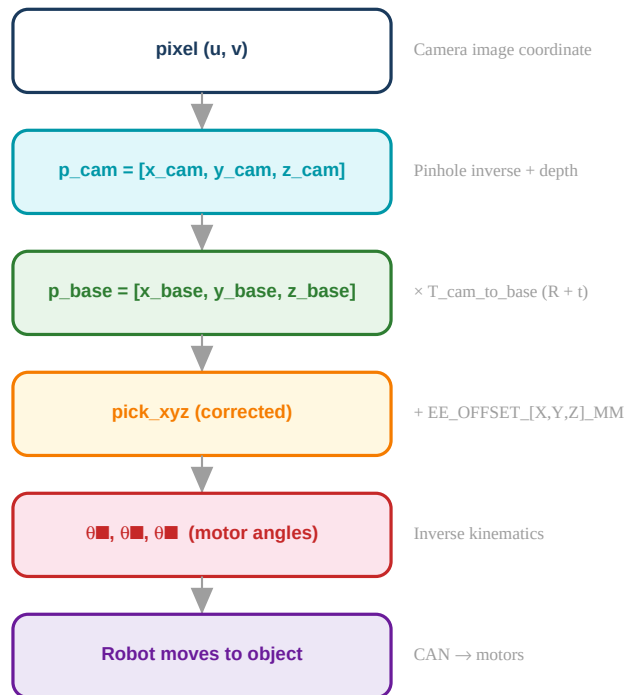


Fig 4 — Full transform chain: pixel → 3D base frame → corrected pick → IK → motor move

5 Tuning Parameters

Parameter	Current Value	What it affects	How to tune
CAM_TY_MM	-300.0 mm	3D y_base coords + overlay Y position	Place obj at Y=0, read printed Y, set = -printed_Y
CAM_TZ_MM	352.0 mm	3D z_base coords (depth from camera)	Measure camera height above belt surface: h_cam + PLACE_Z
EE_OFFSET_X_MM	+50.0 mm	Shifts pick X BEFORE IK	EE overshoots +Xmm → set to -X
EE_OFFSET_Y_MM	+15.0 mm	Shifts pick Y BEFORE IK	EE overshoots +Ymm → set to -Y
WORKSPACE_OVERLAY_Y_OFFSET_MM	0.0 mm	Visual overlay ONLY (no robot effect)	Adjust until green box aligns with physical workspace
ROBOT_SPEED_MM_S	500.0 mm/s	Y prediction for moving belt pick	EE before obj → increase EE after obj → decrease
CONVEYOR_APPROACH_WAIT_SEC	0.3 s	Hold at approach Z before descending	Increase if EE still arrives before object